

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-503: Programming in C

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. fa) What is the use of printf() and scanf) functions? [1] - (b) Name four header files in 'C' programming language. [1]

(a) Determine the hierarchy of - operations and evaluate the " following expression, assuming that -

(b) is a float variable : ; 1 © k=3/24*44+3/8 .

—(d)Whatistheuseoffunctiontoupper()?Explainwithan, -examplecode., 1..|fe)If a = 10, b = 12 and c = 0, then find the values of the following two expressions : .1|

(b) al=688b>5 . fii) 1(a>5 c) [7] () What are the basic data types associated with 'C' programming language? 1 a

(c) What is the output 'of the following | program? : . | ! . maint) GG . | : int x=101, y=105; . ro , . printf ("

(d) What is the difference between ++a | and at++ ? , 1

(e) Describe the difference between = : : and == operators in " . programming. [1]

7 (3) . |

(f) (i) What is the code in while loop that . returns the output of the following code? [1]
include<stdio.h> . | int main() © for (inti=0;i<5;i++) . printf("- " return 0;

Question 2. (a) Write a brief history of ©' programming language. [5] b) Write a 'C' program to determine : the Greatest Common Divisor. (GCD) of two natural numbers. [5]

Question 3. (a) What are the differences between the local variable and the global variable in 'C' with reference to their declaration, visibility and . lifetime? [5]

(a) Write a general program in 'C' to find the sum of the series, x2 | x3 x . X + + —
+ + ---- 3f65!(2n - 1)!andprintitontheoutputscreen.;5

Question 4. ee [3]

Question 4. (a) What are the 'entry-controlled loop and exit-controlled loop? Give . examples. Using a while loop, write

- (b) 'C' program to evaluate the equation $y = x^n$, when n is a non-negative integer. [5]
- (b) Write a general program in 'C' to find the length of a string and count the number of vowels in it.. [5]

Question 5. fa) Write a 'C' program to print the $\circ$ multiplication table. from 1x1 to 12x10. [5]

- (a) What is array? Write the declaration, initialization and memory allocation of a one-dimensional array. 5b) Compute and print the sum of all elements stored in an array using pointer.. [5]

(a) Write a program to print — .

(b) the word MATHEMATICS;

(c) the count of the occurrence — and location of the letter 'A' in that word. ' 5 D2(502) .
(Continued)

Question 7. (a) Write a 'C' program that uses - functions to perform the following' operations using. structure : 5 . fj) Reading a complex number .

(a) Writing a complex number m, fii) Addition of two complex numbers , fiv) Multiplication of two complex ; numbers.

(b) Write a program to read a series of three digit integers from a file named DATA and then write all odd numbers to a file to be called ODD

(c) and all even numbers to a file to be ; called EVEN. [5] ; 7 tk